Hydrogeologic and Public-Records Investigation

Shallow Groundwater, Aquifers, Wetlands, Soils, Drainage, and Geology near 1140 Plett Rd, Cadillac, Michigan

Prepared: August 7, 2026 **Method:** Public records, official government datasets, live GIS/API queries, scientific literature, and lawful records-request planning only. **Status of statements:** Every finding is tagged **[Confirmed]**, **[Probable]**, **[Possible/Inferred]**, or **[Unknown – requires records request]**. Nothing below should be read as asserting facts beyond the cited sources.

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1. Executive Summary

What is known. 1140 Plett Rd sits at ~1,298.5 ft elevation on the north edge of Cadillac, in Haring Charter Township (Section 34, T22N R9W), Wexford County, roughly 950 ft south of where the Clam River crosses Plett Rd and directly across the road from the City of Cadillac Wastewater Treatment Plant (1121 Plett Rd). The area is exceptionally well characterized compared with most rural Michigan sites, because the U.S. Geological Survey published a full hydrogeologic study and groundwater-flow model of this exact watershed — **USGS Scientific Investigations Report 2004-5175 (Hoard & Westjohn, 2005)** — to delineate Cadillac's wellhead protection area. That study documents a thick (600 to >900 ft) sequence of glacial sand and gravel with **four stacked aquifers separated by three clay confining units**, in an ice-block stagnation/outwash landscape. A live query of the State of Michigan's Wellogic database (August 2026) found **519 water wells within 2 miles** of the site and **37 within ~0.5 mile**, all completed in glacial drift ("ice-contact outwash" land system), with well depths of 30–215 ft and **static water levels of roughly 16–93 ft below grade** (clustering ~20–45 ft). The nearest queried log (1118 Plett Rd, 75 ft deep) shows sand from 0–75 ft with a single thin clay bed at 17–20 ft, and a static level of 44 ft.

What is inferred. The site sits on well-to-excessively drained sandy outwash soils (Montcalm-Graycalm complex, non-hydric). The uppermost aquifer is an **unconfined sand aquifer with a water table on the order of 20–45 ft below ground surface at the site** — deeper than typical "shallow groundwater" concerns, but locally variable. Thin clay lenses (like the 17–20 ft bed at 1118 Plett Rd) could seasonally perch small amounts of water, though soils data show no evidence of a seasonal high water table in these map units. Shallow groundwater in this part of the basin likely flows toward the Clam River / basin center, while EGLE's PFAS investigations in the adjacent industrial park state that groundwater flow is "expected to be to the northwest in all aquifers." The nearest well record, taken at face value, yields a water-table elevation ~8 ft above the controlled river/lake base level — consistent with shallow flow toward a gaining Clam River — but a verified elevation/location conflict in that record (Section 5) means no groundwater-elevation math should be trusted until well locations are re-verified against LiDAR.

What remains unknown. National Wetlands Inventory polygons and FEMA flood-zone geometry at the site could not be machine-verified (both federal services blocked automated queries — they work interactively); pre-2000 scanned well logs for Section 34 have not been pulled; drift thickness at the exact parcel is unpublished; the county drain map has not been checked for Haring Township drains; DHD#10 well/septic files, Haring Township minutes, drain commissioner complaint files, and the full EGLE facility file for the WWTP and nearby PFAS Areas of Interest are **not publicly accessible without request**. The April 2026 Cadillac flood (which forced a partially treated WWTP discharge to the Clam River and an ongoing EGLE investigation) is a significant recent hydrologic event whose records are still accumulating.

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2. Location and Jurisdiction Summary

Item	**Value**	**Confidence**
Address	1140 Plett Rd, Cadillac, MI 49601	Confirmed
Coordinates (Census geocoder, address-range interpolated)	44.26464°N, -85.39643°W	Confirmed (±~50–100 ft typical for interpolation; verify against parcel)
Ground elevation at point	1,298.5 ft (USGS 3DEP 1-m DEM, EPQS query)	Confirmed
Parcel ID	2209-34-2208 (2.97 ac; ~9,000 sq ft commercial building, built 1979, former printing company)	Confirmed (listing/assessing sources; verify at BS&A Online)
PLSS	Section 34, T22N, R9W (BLM CadNSDI point query; parcel number encodes it)	Confirmed
Municipality	Haring Charter Township (not City of Cadillac)	Confirmed
County	Wexford County	Confirmed

Item	Value	Confidence
HUC-12	040601020303 "Pleasant Lake–Clam River" (USGS WBD point query)	Confirmed
HUC-10 / HUC-8	0406010203 (Clam River) / 04060102 (Muskegon River basin)	Confirmed
Nearest stream	Clam River, ~950 ft (~290 m) north at the Plett Rd crossing (USGS site 04121241 "Clam River at Plett Road at Cadillac")	Confirmed location; distance approximate
Nearby lakes	Lake Cadillac ~1.5–2 km SW (legal level 1,290.00 ft, Circuit Court Order No. 565); Lake Mitchell beyond, connected by Clam Lake Canal	Confirmed
Adjacent facility	City of Cadillac WWTP, 1121 Plett Rd (advanced tertiary plant, discharges to Clam River; NPDES reissued 11/29/2023; likely permit MI0020257 – probable, verify)	Confirmed / Probable
Nearby investigations	Cadillac Industrial Park PFAS Area of Interest; Wexford-Missaukee Career Tech Center PFAS AOI (9901 E. 13th St — immediate neighborhood); Wexford AOI Self-Sampling Investigation (covers Haring Twp)	Confirmed
Topo quad	Cadillac North, MI 7.5-minute	Confirmed
Key agencies	EGLE (WRD, DWEHD, Remediation/MPART), Wexford Co. Drain	Confirmed

Item	Value	Confidence
	Commissioner, Road Commission, Equalization/GIS, Building Dept., Haring Charter Twp, District Health Department #10, Wexford Conservation District/NRCS, USGS, FEMA, USFWS	

Parcel-boundary confirmation path: BS&A Online (https://bsaonline.com/?uid=553, search "1140 Plett" — free) and Wexford County Equalization/GIS (231-779-9470; interactive GIS is subscription: $250/yr or $20/day; Regrid https://app.regrid.com/us/mi/wexford is a free alternative).

Important geocoding caveat: the coordinate above is interpolated from the road address range, not parcel-centroid. Before building any cross-section, obtain the true parcel geometry from Equalization/BS&A. **[Confirmed limitation]**

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3. Data Inventory Table

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
1	USGS SIR 2004-5175, "Hydrogeology and simulation of ground-water flow in the vicinity of	USGS	https://pubs.usgs.gov/sir/2004/5175/	Report + MODFLOW model	Clam River watershed (96 mi²)	**Anchor document** — aquifer geometry, K values, flow directions, recharge, WHPA	High	Includes Haring Twp; model archive requestable from USGS MI Water Science Center

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
	Cadillac, Michigan" (Hoard & Westjohn, 2005)							
2	Wellogic water-well database (post-1999, digital)	EGLE	https://www.egle.state.mi.us/wellogic ; REST: https://gisagoegle.state.mi.us/arcgis/rest/services/EGLE/DwOpenData/MapServer (Layer 3 wells, Table 4 lithology)	Point DB + lithology table	Statewide	Well depths, SWL, screens, lithology, aquifer type	High for existence; moderate for SWL accuracy (driller-reported, single date)	Live-queried Aug 2026: 519 wells ≤2 mi; 37 wells ≤800 m
3	Scanned well records (pre-2000)	EGLE Well Record Retrieval	https://www.egle.state.mi.us/well-logs/ (search T22N	Scanned PDFs	County/twp/section	Older wells nearest the parcel; 1979 building likely	High	**Not yet pulled — priority action**

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
			R9W Sec 34)			had a well		
4	Water Well Viewer	EGLE	https://www.mcgi.state.mi.us/waterwellviewer/	Web GIS	Statewide	Wells + WHPAs + contamination sites overlay, updated daily	High	Interactive
5	SSURGO / SoilWeb	USDA-NRCS / UC Davis	https://casoilresource.lawr.ucdavis.edu/gmap/ ; Web Soil Survey https://www.nrcs.usda.gov/resources/data-and-reports/web-soil-survey	Polygon + tabular	1:15,840–1:24,000	Drainage class, HSG, water table, hydric rating	High	Site MU: Montcalm-Graycalm complex 0–6% (mukey 190749), live-queried
6	NWI wetlands	USFWS	https://fwsprimary.wim.usgs.gov/wetlands/apps/wetland	Polygon	1:24,000 nominal	Mapped wetlands near Clam River corridor	Moderate (NWI omits many small/forested wetlands)	**REST API blocked to automated client —**

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
			s-mapper/					**check interactively; polygons at site unverified**
7	EGLE Wetlands Map Viewer (incl. MI wetland inventory)	EGLE	https://www.michigan.gov/egle/maps-data/wetlands-map-viewer ; service: https://gisagoegle.state.mi.us/arcgis/rest/services/EGLE/WetlandsMapViewer/MapServer	Web GIS	Statewide	State wetland inventory + hydric soils overlay	Moderate	Not point-verified
8	FEMA NFHL / FIRM panels	FEMA	https://msc.fema.gov/portal/search	Polygon /panels	County	Floodplain of Clam River near site	High if mapped	**Coverage & panel numbers unverified — REST**

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
								blocked; check MSC + Community Status Book
9	NHD / WBD hydrography	USGS	https://hydro.nationalmap.gov/arcgis/rest/services/wbd/MapServer (WBD confirmed working)	Line/polygon	1:24,000	Streams, ditches, HUC boundaries	High	HUC-12 point query succeeded
10	3DEP LiDAR / 1-m DEM (QL2, ~2016 NRCS 30-county MI collection)	USGS / MiSAIL	https://apps.nationalmap.gov/downloader/ ; status: https://gis-michigan.opendata.arcgis.com/datasets/misail-ql2-lidar-status	Raster/point cloud	1 m	Micro-topography, depressions, ditches, wetland basins	High	EPQS 1-m query at site succeeded (1,298.53 ft)

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
11	Farrand & Bell 1982 Quaternary Geology of Michigan	MGS/MDNR (digitized MNFI 1998)	https://wmich.edu/sites/default/files/attachments/u263/2014/1982_Quaternary_Geology_Map.pdf	Map (1:500,000)	Statewide	Surficial units: outwash, coarse-till end moraines around Cadillac basin	Moderate at parcel scale	USGS 2004 reinterprets moraines as ice-contact sand/gravel
12	GWIM statewide groundwater maps (2005–06)	MSU/EGLE/USGS	https://www.egr.msu.edu/igw/GWIM%20Figure%20Webpage/ ; "Aquifer Characteristics of Glacial Drift": https://gis-egle.hub.arcgis.com/maps/egle::aquifer-characteristics-of-glacial-drift	Raster/vector	Statewide	Water-table depth, recharge, transmissivity, land systems	Moderate	Derived from 484k Wellogic records

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
13	USGS OFR 2007-1236 county hydrogeologic summaries	USGS	https://pubs.usgs.gov/of/2007/1236/pdf/OFR2007-1236.pdf	Report	County	Wexford County chapter — aquifer summary + bibliography	High	
14	Holtschlag 1996 recharge estimates (OFR 96-593)	USGS	https://pubs.usgs.gov/publication/ofr96593	Report/ grid	Lower Peninsula	~10 in/yr recharge near Cadillac; model used 15 in/yr	High	
15	NWIS groundwater sites, Wexford Co.	USGS	https://waterservices.usgs.gov/nwis/site/?format=rdb&countyCd=26165&siteType=GW&siteStatus=all	Point	County	9 GW sites; incl. 441554085230601 "22N 09W 33B01 (Haring 1)" ~1 mi west of site	High	Water-level time series now via https://api.waterdata.usgs.gov (legacy API retired) — records not yet retrieved
16	NWIS surface-water sites,	USGS	Site 04121241 (Clam	Point	Local	Historic discharge measur	High	No active continu

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
	Clam River		R. at Plett Rd; 10 discrete measurements 1975–2001) and 8 other Clam R./tributary sites			ements bracketing the site		ous gage
17	GeoWebFace (rebuilt July 2025)	EGLE	https://www.egle.state.mi.us/geowebface/	Web GIS	Statewide	Oil/gas wells (deep lithology), bedrock/Quaternary layers	High	
18	EGLE MiEnviro Portal	EGLE	https://mienviro.michigan.gov (Site Map Explorer, public)	Documents DB	Statewide	WWTP permits/inspections, AOI docs	Moderate (incomplete pre-2015; drafts excluded)	Full file requires FOIA
19	PFAS AOI pages (Cadillac Industrial Park;	EGLE/MPART	https://www.michigan.gov/pfasresponse/investigations/	Reports/maps	Local	Flow-direction statements, residential well samplin	High for what's posted	Consultant hydro reports mostly not

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
	Wexford-Missaukee CTC; Wexford Self-Sampling)		sites-aoi/wexford-county/			g in Haring Twp		posted — FOIA
20	County Drain Map	Wexford Co. Drain Commissioner	https://wexfordcounty.org/wp-content/uploads/2026/07/County_Drain_Map.pdf	PDF map	County	Which of 17 county drains (if any) are near the site	Unknown until read	**Not yet reviewed**
21	Historic aerial photos (1930s–2000s)	MSU RS&GIS archive	https://rsgis.msu.edu/imagery-archive/information ; slider: https://apps.gis.msu.edu/aerial-imagery/slider/	Imagery	Frame	Historical drainage/wetland/land-use changes	High	~$30/frame ordering; also historicaerials.com, Vintage Aerial, Wexford Co. Historical Society (1961+)
22	Historic USGS topos (Cadilla	USGS topoView	https://ngmdb.usgs.gov/topovie	Scanned maps	1:24,000 / 1:62,500	Historic wetlands, streams, ditches	High	Edition years not yet enumerated

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
	c North quad)		w/viewer/					
23	Wexford Co. soil survey (modern, 1980s-era; historical 1909 Geib survey)	NRCS / Wexford Conservation District	Paper copies free at Conservation District; 1909 scan via NRCS archived manuscripts	Report	County	Manuscript descriptions, water-table notes	Moderate	Exact modern publication year unverified
24	City of Cadillac water/wastewater fact sheet (Apr 2025), PFAS hub, AOI FAQ	City of Cadillac	https://www.cadillac-mi.net/DocumentCenter/View/3000/... ; https://www.cadillac-mi.net/PFAS	PDFs	Local	WWTP operations, aquifer framing, well-field history	High	
25	County drain/lake-level records (Order No. 565; Clam River Dam	Drain Commissioner	https://wexfordcounty.org/?page_id=469	Office records	Local	Controls base-level of Lake Cadillac (1,290.00 ft)	High	Operational records require request

#	Dataset / record	Custodian	Access path	Type	Scale	Relevance	Reliability	Notes
	operation)							

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4. Hydrogeologic Findings

4.1 Regional framework — the single best source

USGS SIR 2004-5175 (Hoard & Westjohn, 2005) characterized the Clam River watershed, which contains the site, for Cadillac's wellhead protection program. Its key published values **[all Confirmed, primary source]**:

- Glacial deposits 600 to >900 ft thick (citing Westjohn et al., 1994); no bedrock aquifers are used anywhere in the watershed.
- **Four stacked glacial aquifers, three clay confining units**, in ft above NGVD 29: shallow aquifer ~1,300–1,278; intermediate aquifer 1 ~1,273–1,139; intermediate aquifer 2 ~1,135–1,073; deep aquifer ~1,050–700.
- Aquifer-test hydraulic conductivities: shallow aquifer Kh = 129.6 ft/d (screens 20–35 ft); intermediate 43.9–163.0 ft/d; **Haring Township well field (deep aquifer, 1986 test): Kh = 109.5 ft/d, screen 228–268 ft**; deep clay confining unit Kv = 0.031 ft/d (leaky).
- Flow: shallow/intermediate groundwater flows from the NW and SE highlands **toward the basin center (the lakes/river system)**; deep-aquifer regional flow is **SE** → **NW**. Lakes held at ~1,288.9–1,290.0 ft.
- Water balance: precipitation ~32.5 in/yr (older normal; 1991–2020 normals ~36 in/yr), effective recharge ~10 in/yr (Holtschlag), model-calibrated recharge 15 in/yr.
- Landscape reinterpretation: the "moraines" ringing the basin are largely **coarse sand and gravel**; the basin is an **ice-block stagnation/collapse complex** — Lakes Cadillac/Mitchell and associated wetlands occupy collapsed outwash over buried ice.

With site grade at ~1,298.5 ft, the site surface sits at the very top of the mapped **shallow aquifer interval (1,300–1,278 ft)** — i.e., the uppermost ~20 ft of saturated sand mapped by USGS lies roughly 0–20+ ft below the parcel, **where saturated**. Local static levels (below) indicate the actual water table at the site is deeper than the top of that mapped interval. **[Inferred, medium confidence]**

4.2 Groundwater and wells (live Wellogic query, Aug 2026)

- **519 wells within 2 miles** of the site; **37 within ~800 m (0.5 mi)**. **[Confirmed]**

- Depths within 800 m: 30–215 ft, most 40–110 ft. Static water levels: **16–93 ft below grade, clustering ~20–45 ft**. All sampled records: aquifer type = DRIFT; land system = "ICE-CONTACT OUTWASH." **[Confirmed]**
- Nearest queried log — **1118 Plett Rd** (WELLID 83000000974, Sec 34 T22N R9W, drilled 1996): 75 ft deep, 2-inch driven well, screen 71–75 ft, **SWL 44 ft**. Lithology: **Sand 0–17 ft; Clay 17–20 ft; Sand 20–75 ft. [Confirmed]**
- Nearby: 9052 E. 13th St — 110 ft deep, screen 106–110 ft, SWL 40 ft. **[Confirmed]**
- USGS observation well **441554085230601 "22N 09W 33B01 (Haring 1)"** sits ~1 mile west (Section 33). Water-level period of record not yet retrieved (legacy NWIS groundwater-levels API retired; use https://api.waterdata.usgs.gov). **[Confirmed site; records = gap]**
- Flowing/artesian wells: none observed in the sampled records near the site; the FLOWING field exists in Wellogic and should be checked systematically. **[Possible gap]**

Interpretation: the uppermost saturated zone at the site is an **unconfined sand aquifer with a water table plausibly ~20–45 ft below grade** — not "shallow" in the wet-basement sense, but shallow in aquifer terms. Wells screened deeper (100–215 ft) tap the intermediate aquifers; their SWLs (some reported to 93 ft) may reflect different (lower) heads, which — if verified against surveyed elevations — would indicate **downward vertical gradients, i.e., the site is in a recharge area**. **[Inferred, medium confidence]**

4.3 Aquifers — summary

- **Surficial/unconfined aquifer:** ice-contact outwash sand, high Kh (~130 ft/d tested nearby), hosting domestic wells at 30–110 ft. **[Confirmed presence; Probable properties at parcel]**
- **Perched aquifers:** thin clay beds exist (17–20 ft at 1118 Plett Rd) and could locally perch water seasonally; soils show no seasonal high water table, so any perching is likely thin, discontinuous, and transient. **[Possible]**
- **Confined/semi-confined aquifers:** intermediate aquifers 1–2 and the deep aquifer (Cadillac and Haring Twp municipal source, screens ~228–430 ft) beneath clay units. **[Confirmed regionally]**
- **Wetland-connected groundwater systems:** the Clam River corridor ~950 ft north and the collapsed-outwash wetlands around Lakes Cadillac/Mitchell are the likely discharge zones for the shallow system. **[Probable]**

4.4 Soils (SoilWeb/SSURGO live query at the coordinates)

Site map unit: **mukey 190749 — Montcalm-Graycalm complex, 0–6% slopes [Confirmed]**:

- Montcalm (50%) — sandy Spodosol (Alfic Haplorthods), **well drained, non-hydric**, parent material sandy drift; available water ~14 cm.
- Graycalm (35%) — sandy Entisol (Lamellic Udipsamments), somewhat excessively drained.

- Minor: Rubicon (8%), Manistee (7%). Adjacent units within ~200 m: Grattan, Coloma, Rubicon — all deep, excessively drained sands.
- Hydrologic soil group A (typical for these series — **[Probable]**; exact SSURGO hydgrp/musym not machine-retrieved because the Soil Data Access API is POST-only; confirm in Web Soil Survey).
- **Implications:** rapid infiltration, high recharge, negligible runoff, no hydric indicators, no listed seasonal high water table, no flooding/ponding classes expected. These soils argue **against** near-surface (0–6 ft) saturation on the parcel itself and **for** a recharge-dominated setting. Septic limitations would be "poor filter" (too permeable), not wetness. **[Inferred, medium-high confidence]**

4.5 Geology and geomorphology

- Surficial: outwash sand/gravel basin ringed by ridges mapped as coarse-till end moraines (Farrand & Bell 1982) but reinterpreted by USGS (2004) as ice-contact sand/gravel of a stagnation complex. Site landform: gently sloping outwash/ice-contact plain between the Clam River (north) and the urban edge (south). **[Confirmed regionally; Probable at parcel]**
- Bedrock: Mississippian–Pennsylvanian sandstone/shale/carbonate/evaporite at great depth (600–900+ ft); unit directly beneath the site (Michigan Fm vs. Marshall Ss) **[Unknown detail]**; irrelevant to shallow hydrology here.
- Landform risks to check on LiDAR: kettle depressions (collapsed outwash), abandoned channel scars along the Clam River, made-land/fill on the commercial parcel (built 1979). **[Possible — LiDAR task]**

4.6 Wetlands

- NWI polygons at/near the site: **unverified** — the USFWS wetlands REST service returned empty responses to automated queries; use the Wetlands Mapper interactively. Riparian wetlands along the Clam River corridor ~950 ft north are plausible but unverified. **[Unknown/Probable]**
- EGLE Wetlands Map Viewer (includes the state inventory and hydric-soil layers): not point-verified. **[Gap]**
- Note: Michigan's wetland inventory maps are non-regulatory; an on-site delineation governs jurisdictional status.

4.7 Drainage and surface water

- Clam River: outlet of Lake Cadillac (legal level 1,290.00 ft, Order No. 565; dam operated by the Drain Commissioner), flows NE through the city, crosses Plett Rd ~950 ft north of the site, continues to the Muskegon River (51-mile tributary). Designated cold-water trout stream some miles downstream of the WWTP discharge. **[Confirmed]**

- USGS measurement sites bracket the reach (04121235–04121247, incl. 04121241 at Plett Rd, 10 discrete measurements 1975–2001); **no active continuous gage**. **[Confirmed]**
- County drains: the Drain Commissioner maintains **17 established county drains**; whether any lie in Haring Twp Section 34 is unknown until the County Drain Map PDF is reviewed / office is asked. **[Unknown — action item]**
- Road drainage: Plett Rd jurisdiction (city vs. county road) unverified; culvert and ditch records sit with the Road Commission (and/or City). **[Unknown — request]**
- **April 2026 flood:** historic flooding in Cadillac shut down a sanitary sewer station, forced the WWTP to bypass tertiary treatment to the Clam River, and triggered an EGLE investigation (violations issued for reporting deficiencies; city response due July 31, 2026). This event is direct evidence of the local system's high-water behavior and will have generated records (EGLE, city, county EM) worth requesting. **[Confirmed]**

4.8 Floodplain

- FEMA NFHL coverage, zone, and panel numbers for this reach of the Clam River: **unverified** (automated NFHL queries blocked). Check https://msc.fema.gov/portal/search for the address and FEMA's Community Status Book for Haring Charter Township NFIP participation. Site grade ~1,298.5 ft vs. river base-level ~1,290 ft suggests the parcel itself is above the riverine floodplain, but the April 2026 flood shows the local system can exceed expectations. **[Unknown mapping; Inferred low flood exposure at parcel, low-medium confidence]**

4.9 Historical changes

- Building constructed 1979 (former printing company); site history before 1979 unknown — check MSU aerial archive (1938+), historic topos, and 1961+ county historical society aerials for prior wetlands, borrow pits, or drainage alterations. **[Gap]**
- The WWTP across the road dates to 1962 (expanded 1975/1989/1996; upgraded 2003/2007) — its construction records would document mid-century subsurface conditions on Plett Rd. **[Confirmed facility history; records = request]**

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5. Well-Log Analysis Plan

Data exist to do this properly — 519 wells within 2 miles, with depth/SWL/screen/lithology fields, plus a lithology table keyed by WELLID.

Starter table (from live queries; expand to all 37+ wells within 0.5 mi):

Well	Distance	Ground elev.*	Total depth	SWL	GW elev.*	Lithology highlights	Screen	Aquifer	Confidence
1118 Plett Rd (830000097 4, 1996)	~500 ft	**1,342 ft (ELEV_DEM field — verified by direct query 8/7/2026, but suspect; see anomaly note)**	75 ft	44 ft	~1,298 ft (if ELEV_DEM correct)	Sand 0–17; **clay 17–20**; sand 20–75	71–75 ft	Drift, unconfined	High (log), **Low (elev./location)**
9052 E. 13th St	~0.3–0.5 mi	TBD	110 ft	40 ft	TBD	TBD (pull lithology table)	106–110 ft	Drift	High (log)
USGS 441554085230601 (Haring 1, Sec 33)	~1 mi W	TBD	TBD	time series	TBD	TBD	TBD	TBD	Records not yet pulled

*Use the Wellogic ELEV_DEM field as a first pass, then re-extract elevations from the 1-m 3DEP DEM at each well's lat/lon — driller-reported locations can be off by hundreds of feet, so grade each well A/B/C for location reliability before contouring.

Workflow:

1. Query Layer 3 (wells) within 3,218 m of the site with all fields; export to CSV.
2. Join Table 4 (lithology) on WELLID.
3. Compute GW elevation = ELEV_DEM − SWL; flag wells with screens >100 ft as potentially representing deeper heads (do not mix into the water-table map).
4. Pull pre-2000 scanned logs for T22N R9W Sections 27, 28, 33, 34, 35 from https://www.egle.state.mi.us/well-logs/ and hand-digitize.
5. Build: well-location map; depth histogram; SWL table; lithology correlation table; preliminary water-table elevation contours (shallow wells only); shallow-aquifer thickness estimate (top of first laterally persistent clay from lithologies); two cross-sections — (A) along the topographic gradient S→N from the urban edge through the site to the Clam River; (B) W→E through the site toward the USGS Haring 1 well.
6. Compare contoured water-table elevations against Clam River stage (~1,288–1,290 ft near the crossing, needs survey/LiDAR water-surface extraction) and Lake Cadillac's 1,290.00 ft legal level.

Key anomaly to resolve (verified by direct re-query, 8/7/2026): the Wellogic record for the 1118 Plett Rd well reports **ELEV_DEM = 1,342 ft**, yet its recorded coordinates (~44.267, -85.396) plot near the Clam River crossing, where the 1-m 3DEP surface is in the low 1,290s, and the site itself (44.2646) is at 1,298.5 ft. The three numbers cannot all be right. If ELEV_DEM is correct, groundwater elevation ≈ 1,342 − 44 = **~1,298 ft**, about 8 ft above the controlled river/lake base level — consistent with shallow flow toward a gaining Clam River. If the recorded location/elevation is wrong (common in Wellogic; locations are driller-reported), the computed head is unreliable in either direction. **Resolution path:** re-extract ground elevation from the 3DEP 1-m DEM at each well's best-verified location (match WELL_ADDR to the parcel, not the stored lat/lon) before computing any groundwater elevations, and grade every well A/B/C for location reliability. Do not contour until this is done. **[Open question — high priority]**

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6. GIS Workflow (step-by-step)

1. **Geocode & parcel:** confirm parcel 2209-34-2208 geometry via BS&A Online / Equalization (or Regrid export). Reproject everything to NAD83 Michigan GeoRef or UTM 16N; use NAVD88 vertical.
2. **DEM:** download 1-m 3DEP DEM tiles for a 2-mile buffer from https://apps.nationalmap.gov/downloader/ (QL2 LiDAR, ~2016). Also grab the LAS point cloud if micro-features matter.
3. **Hydro-condition & derive:** fill/breach the DEM; compute flow direction, flow accumulation, and closed depressions (Fill-minus-DEM); extract drainage divides and swale/ditch alignments; extract Clam River water-surface elevation profile from LiDAR returns.

4. **Hydrography:** overlay NHD flowlines/waterbodies and the WBD HUC-12 (040601020303) boundary.
5. **Wetlands:** overlay NWI (interactive download via Wetlands Mapper) and EGLE Wetlands Map Viewer layers; note this is the step that fills the current wetlands gap.
6. **Soils:** overlay SSURGO (Web Soil Survey AOI export or gSSURGO); symbolize by drainage class, hydric rating, HSG, and depth-to-water-table attribute (soilmu_a + component/chorizon tables).
7. **Geology:** overlay Farrand & Bell Quaternary polygons (BTAA/MNFI digitized version) and GWIM land-system and aquifer-characteristics layers.
8. **Wells:** pull EGLE DwOpenData Layer 3 + lithology Table 4 (REST URLs in Section 3); add USGS NWIS GW sites; add scanned-log wells hand-located from T/R/S descriptions.
9. **Water levels:** compute groundwater elevations; contour water table using shallow wells only; estimate the local hydraulic gradient and flow direction vector at the site.
10. **Recharge/discharge:** classify recharge (sandy uplands, HSG A, no wetlands) vs. discharge (river corridor, NWI polygons, muck soils) zones; compare with GWIM recharge grid (~10–15 in/yr).
11. **Floodplain:** add FEMA NFHL (interactive download) once coverage is confirmed.
12. **Cross-sections:** construct sections A (S→N to the Clam River) and B (W→E to the Haring 1 well) in QGIS profile tool or Excel from the lithology table; correlate clay beds; distinguish aquifer/aquitard.
13. **CSM assembly:** integrate into the conceptual model (Section 7) and map key uncertainties.
14. **Optional verification:** request the USGS MODFLOW model archive for SIR 2004-5175 from the Michigan Water Science Center and compare simulated heads at the site to your contours.

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7. Preliminary Conceptual Site Model (CSM)

Setting: sandy ice-contact outwash plain at ~1,298.5 ft, on the northern rim of the collapsed ice-block basin that holds Cadillac and its lakes; Clam River ~950 ft north flowing NE; base level controlled at 1,290.00 ft at the Lake Cadillac dam.

CSM element	Interpretation	Status
Shallow geologic units	Fine-to-coarse ice-contact outwash sand, ≥75–110 ft thick before first major clay; thin discontinuous clay lenses (e.g., 17–20 ft bgs nearby)	**Probable** (nearby logs; parcel-specific log absent)

CSM element	Interpretation	Status
Groundwater occurrence	Unconfined water table on the order of 20–45 ft bgs at the parcel; saturated thickness of shallow aquifer tens of feet	**Probable**
Perched water	Possible thin, seasonal perching on clay lenses; no soil-based evidence of near-surface (0–6 ft) saturation	**Possible**
Aquifer material	High-K outwash sand (tested Kh ≈ 44–163 ft/d in the watershed; 129.6 ft/d for the shallow aquifer)	**Confirmed regionally / Probable locally**
Restrictive layers	Three regional clay confining units at depth; uppermost significant clay locally variable	**Confirmed regionally**
Recharge	Direct precipitation infiltration through HSG-A sands, ~10–15 in/yr; parcel is in a recharge zone	**Probable**
Discharge	Clam River corridor and lake/wetland system (basin center); regional deep flow exits NW	**Probable (shallow) / Confirmed (deep, per USGS & EGLE statements)**
GW–SW interaction	River likely gains from the shallow aquifer along this reach (face-value nearest-well head ≈ 1,298 ft vs. river base ~1,290 ft), but the verified elevation/location conflict in that record (Sec. 5) leaves this unproven	**Probable, pending well-location verification**

CSM element	Interpretation	Status
Role of drains/ditches	County drain presence unknown; road ditches on Plett Rd likely convey runoff to the river; WWTP outfall is a point discharge to the river	**Unknown/Probable**
Vertical gradients	Likely downward (recharge setting); deep aquifer semi-confined with leaky clay (Kv 0.031 ft/d)	**Probable**
Anthropogenic influences	WWTP operations across the road (surface discharger — no evidence of lagoons/infiltration basins); PFAS AOIs in the immediate area with stated NW flow; municipal pumping relocated to Crosby Rd/44 Rd fields; April 2026 flood event	**Confirmed**

Key uncertainties: (1) water-table elevation and flow direction at the parcel itself; (2) reconciling "flow toward basin center" (USGS shallow) with "flow NW in all aquifers" (EGLE AOI statements — likely reflecting the industrial-park area north/west of the lakes, but the CTC AOI, in this immediate neighborhood, says flow is "unknown but may potentially be to the northwest towards the Clam River"); (3) continuity of the shallow clay lens; (4) wetlands/floodplain geometry (unverified layers); (5) any parcel-specific well/septic history.

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8. Data Gap Matrix

Data gap	Why it matters	Best source	How to obtain	Priority	Difficulty	Confidence impact
Parcel boundary + any well/septic	Anchors all mapping; 1979 building likely had	BS&A/Equalization; DHD#10; EGLE	Free lookup; FOIA to DHD#10; scanned-lo	**1 — Highest**	Low	High

Data gap	Why it matters	Best source	How to obtain	Priority	Difficulty	Confidence impact
on the parcel	well/septic before any municipal service	scanned logs	g search T22N R9W S34			
Pre-2000 scanned well logs, Secs 27/28/33/34/35	Doubles+ the lithology dataset; oldest wells were shallowest	EGLE Well Record Retrieval	https://www.egle.state.mi.us/well-logs/ (self-serve)	**1**	Low	High
NWI / state wetland polygons near site	Discharge-zone mapping; wetland hydrology indicators	USFWS Wetlands Mapper; EGLE viewer	Interactive query/download (automated API blocked)	**1**	Low	Medium
FEMA NFHL zone/panels	Flood exposure; floodplain = discharge zone proxy	FEMA MSC	Address search msc.fema.gov	**1**	Low	Medium
Water-table elevation anomaly (river vs. SWLs)	Determines flow direction & GW-SW interaction	LiDAR water surface + surveyed well elevations + new SWL measurements	GIS work + possibly field verification	**1**	Medium	High

Data gap	Why it matters	Best source	How to obtain	Priority	Difficulty	Confidence impact
County drain presence in Haring S34	Drains alter shallow flow and legal drainage	Drain Commissioner	Read County_Drain_Map.pdf; call/records request	2	Low	Medium
USGS Haring 1 obs. well water-level record	Only long-term head record within 1 mi	USGS api.waterdata.usgs.gov	API pull (new OGC endpoint)	2	Low	Medium
Consultant hydro reports (industrial park plumes; CTC & Cadillac AOIs; AECOM work plans)	Contain measured water tables, flow maps, cross-sections for the immediate area	EGLE Remediation/MPART files; EPA Superfund records (Kysor, Northernaire)	EGLE FOIA; EPA SEMS/records request	**1**	Medium	**Very high**
WWTP construction/geotech records (1962–2007) & NPDES file	Borings on Plett Rd; any groundwater notes; verify permit MI0020257	EGLE WRD via MiEnviro + FOIA; City of Cadillac	MiEnviro search; FOIA both	2	Medium	Medium-High
April 2026 flood records	High-water marks, groundwater response,	EGLE, City, County EM,	FOIA/requests	2	Medium	Medium

Data gap	Why it matters	Best source	How to obtain	Priority	Difficulty	Confidence impact
	sewer/WWTP performance	Cadillac News				
Drift thickness / bedrock surface at site	Completes deep framework (academic here)	MGS/WMU statewide maps; GeoWebFace oil-gas logs	Purchase/request MGS sheets; GeoWebFace	3	Medium	Low
Historic aerials & topo editions	Pre-1979 land use, drainage alterations	MSU RS&GIS; topoView; county historical society	Order frames; free downloads	2	Low	Medium
Wexford soil survey manuscript (modern + 1909)	Narrative water-table and drainage notes	Conservation District (free paper); NRCS archives	Pickup/request	3	Low	Low-Medium
Haring Twp minutes/permits mentioning Plett Rd	Drainage complaints, development conditions	Township clerk	Records request (not web-indexed)	2	Medium	Medium

9. Public-Records Request Plan

Agency	Department	Records to request	Reason	Priority	Contact path
Michigan EGLE	FOIA Request Center (WRD; Remediation & Redevelopment/MPART; DWEHD)	See Draft A below — AOI/plume hydro reports, WWTP file, well records	Best-quality measured groundwater data near site	**1**	https://www.michigan.gov/egle/contact/foia (online portal)
District Health Dept. #10	Environmental Health	Well & septic permits, soil evaluations/perc tests for 1140 Plett Rd + adjacent parcels	Parcel-specific soils/water-table observations	**1**	foia@dhd10.org; 521 Cobb St, Cadillac; 231-775-9942
Wexford County	Drain Commissioner	Drain files, maps, complaints — Haring Twp	Legal drains & drainage complaints alter shallow flow	2	draincom@wexfordcounty.org; 231-779-9115
Wexford County	Road Commission	Culvert/ditch records, Plett Rd drainage	Road drainage controls local runoff paths	2	admin@wexfordcrc.org; 231-775-9731
Wexford County	Equalization/ GIS + FOIA coordinator	Parcel geometry, orthos, any GIS drainage layers	Base mapping	2	equalization@wexfordcounty.org; foia@wexfordcounty.org

Agency	Department	Records to request	Reason	Priority	Contact path
Wexford County	Building/Zoning	Building & soil-erosion permits for 1140/1121 Plett Rd	Site plans, grading, geotech	3	401 N. Lake St, Cadillac
Haring Charter Township	Clerk	Minutes/permits/complaints referencing Plett Rd, drainage, wells, flooding	Local knowledge not web-indexed	2	505 Bell Ave, Cadillac; 231-775-8822
City of Cadillac	Utilities/Engineering	WWTP construction/geotech borings; April 2026 flood records	Subsurface data on Plett Rd	2	via City Clerk FOIA
Wexford Conservation District / NRCS	—	County soil survey copies; any drainage/wetland files	Manuscript water-table notes	3	7192 E. 34 Road, Cadillac; 231-775-7681
EPA Region 5	Superfund records	Kysor Industrial / Northernaire site files (hydro reports, MW logs)	Measured local hydrogeology	2	EPA SEMS / regional records center
USGS MI Water Science Center	—	SIR 2004-5175 MODFLOW archive;	Model heads at site	2	https://www.usgs.gov/centers/upper-midwest-water-

Agency	Department	Records to request	Reason	Priority	Contact path
		Haring 1 well record			science-center

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10. Draft Records-Request Language

A. EGLE (submit via EGLE FOIA Request Center):

> Pursuant to the Michigan Freedom of Information Act, MCL 15.231 et seq., I request copies of the following records concerning Section 34, T22N, R9W, Haring Charter Township, Wexford County (vicinity of 1140 and 1121 Plett Road, Cadillac, MI 49601), and the surrounding one-mile radius: (1) All hydrogeologic reports, monitoring-well logs, boring logs, groundwater elevation/potentiometric maps, water-table measurements, and cross-sections in the files of the Remediation and Redevelopment Division and MPART for the Cadillac Industrial Park Area of Interest, the Wexford-Missaukee Career Technical Center Area of Interest, and the Wexford Area of Interest Self-Sampling Investigation, including consultant work plans and reports (including but not limited to AECOM documents); (2) Water Resources Division files for the City of Cadillac WWTP (1121 Plett Rd; NPDES permit believed to be MI0020257) and the Haring Township WWTP, including permit applications, basis-of-design documents, geotechnical/soil borings, groundwater discharge evaluations, hydrogeologic studies, and compliance/inspection records, including records related to the April 2026 flood bypass event; (3) Any water well records, abandoned-well records, or well-plugging records for T22N R9W Sections 27, 28, 33, 34, and 35 not available through Wellogic or the scanned well-record retrieval system; (4) Any wetland permit applications, wetland delineations, floodplain determinations, or Part 301/303 files referencing Plett Road, Haring Township. Please provide records electronically. If fees will exceed $50, please contact me first. I request a fee waiver as [state basis, if any].

B. District Health Department #10 (foia@dhd10.org):

> Under the Michigan FOIA, I request copies of all well permits, well records, septic/sewage permits, soil evaluations, percolation tests, site evaluations, and related inspection records for the parcel at 1140 Plett Rd, Cadillac, MI 49601 (Parcel No. 2209-34-2208, Haring Charter Township), and for adjacent parcels fronting Plett Road between 13th Street and the Clam River, from 1960 to present.

Please include records indexed under prior owner names or older addresses if the address has changed.

C. Wexford County Drain Commissioner (draincom@wexfordcounty.org):

I request copies of: (1) the current county drain map and the legal descriptions/route maps of any established county drain located wholly or partly in Haring Charter Township; (2) all maintenance, inspection, or petition files for such drains from 1990 to present; (3) any drainage complaints, correspondence, or investigation records referencing Plett Road, 13th Street, or Section 34 of Haring Township; and (4) records of Clam River Dam operation and Lake Cadillac/Mitchell lake-level management (Court Order No. 565) for 2020–present, including the April 2026 flood period.

D. Wexford County Road Commission (admin@wexfordcrc.org):

I request records showing: (1) the road jurisdiction of Plett Road between 13th Street and Boon Road; (2) culvert inventories, culvert or driveway permits, and ditch maintenance/cleaning records for that segment and its intersecting streets from 1990 to present; and (3) any drainage complaints or flood-damage records for Plett Road, including the April 2026 flood.

E. Wexford County FOIA coordinator (foia@wexfordcounty.org):

I request: (1) the current GIS parcel boundary (shapefile or GeoJSON) and assessing record card for Parcel 2209-34-2208 (1140 Plett Rd, Haring Township); (2) any countywide GIS layers depicting drains, culverts, watercourses, wetlands, or floodplains; and (3) building permits, soil erosion and sedimentation control permits, and associated site plans for 1140 Plett Rd from 1975 to present.

F. Haring Charter Township Clerk (505 Bell Ave; 231-775-8822):

I request: (1) board and planning commission minutes from 1975 to present that reference Plett Road, drainage, flooding, wetlands, wells, or the wastewater treatment plant; (2) any zoning or land-use permits, site plans, or variances for 1140 Plett Rd; and (3) any township drainage studies or infrastructure plans covering Section 34. If minutes are not indexed, I am glad to review them at the office by appointment.

G. Wexford Conservation District / NRCS (7192 E. 34 Rd):

Request (informal): a copy of the Wexford County soil survey (modern and any 1909 Geib survey material), and any conservation-planning, drainage, or wetland files concerning the Plett Road / lower Clam River area.

H. USGS Upper Midwest Water Science Center (informal request, not FOIA):

Request the model archive (input files, calibration heads) for SIR 2004-5175 and the complete water-level record for observation well 441554085230601 ("22N 09W 33B01, Haring 1").

I. EPA Region 5 (records request):

Request Superfund site file excerpts for Kysor Industrial Corp. and Northernaire Plating (Cadillac): remedial investigation reports, monitoring-well boring logs, groundwater contour maps, and any 2021–2025 PFAS sampling of site monitoring wells.

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11. Confidence and Uncertainty Summary

Finding	**Rating**	**Basis**
Site location, PLSS, HUC-12, elevation	**High**	Direct API queries (Census, BLM, USGS WBD, EPQS)
Four-aquifer glacial framework, 600–900+ ft drift, no bedrock aquifer use	**High**	USGS SIR 2004-5175 (primary)
Uppermost aquifer = unconfined ice-contact outwash sand; domestic wells 30–110 ft; SWL ~20–45 ft nearby	**High** (data) / **Medium** (extrapolation to parcel)	Live Wellogic queries; no log on the parcel itself
Soils well/excessively drained, non-hydric, HSG A; recharge setting	**Medium-High**	SoilWeb component data; SSURGO tabular details not machine-verified
Shallow flow toward Clam River/basin center	**Medium**	USGS regional statement; contradicted in part by EGLE "NW in all aquifers" AOI

Finding	Rating	Basis
		statement and the SWL/river-elevation anomaly — unresolved
Perched water on thin clay lenses	**Low/Possible**	Single nearby log; no soil evidence
Wetland presence/absence near site	**Unknown**	NWI/EGLE layers not verified (services blocked to automated client)
FEMA flood zone at site	**Unknown** (parcel likely above riverine floodplain — low-medium confidence inference)	NFHL unverified; grade 8–9 ft above base level
County drain influence	**Unknown — requires records request**	Drain map not yet reviewed
Parcel-specific well/septic history	**Unknown — requires records request**	DHD#10 / scanned logs
WWTP as surface discharger (no lagoons/infiltration)	**Medium-High**	City fact sheet; absence of Part 22 evidence is not proof — verify in MiEnviro/FOIA

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12. Source List (accessed August 7, 2026)

Primary technical: USGS SIR 2004-5175 (https://pubs.usgs.gov/sir/2004/5175/); USGS OFR 2007-1236 (https://pubs.usgs.gov/of/2007/1236/pdf/OFR2007-1236.pdf); Holtschlag OFR 96-593 (https://pubs.usgs.gov/publication/ofr96593); Westjohn et al. WRIR 93-4152 & PP 1418 (citations); Farrand & Bell 1982 map (https://wmich.edu/sites/default/files/attachments/u263/2014/1982_Quaternary_Geology_Map.pdf); GWIM figures (https://www.egr.msu.edu/igw/GWIM%20Figure%20Webpage/); Leverett & Taylor 1915 Monograph 53; Stewart 1948 MSU thesis & Thomson 1969 MTU thesis (citations, full text not located online). **Live data queries:** Census geocoder; BLM CadNSDI PLSS REST; USGS WBD REST (HUC-12); USGS EPQS (elevation); USGS NWIS site services (GW & SW site lists; https://waterservices.usgs.gov/nwis/site/?format=rdb&countyCd=26165&siteType=GW&siteStatus=all); EGLE DwOpenData REST (wells + lithology;

https://gisagoegle.state.mi.us/arcgis/rest/services/EGLE/DwOpenData/MapServer); SoilWeb API (https://casoilresource.lawr.ucdavis.edu/api/landPKS.php?q=spatial&lon=-85.39643&lat=44.26464). **State/local:** EGLE Wellogic (https://www.egle.state.mi.us/wellogic); scanned well records (https://www.egle.state.mi.us/well-logs/); Water Well Viewer (https://www.mcgi.state.mi.us/waterwellviewer/); GeoWebFace (https://www.egle.state.mi.us/geowebface/); EGLE Wetlands Map Viewer (https://www.michigan.gov/egle/maps-data/wetlands-map-viewer); MiEnviro Portal (https://mienviro.michigan.gov); EGLE FOIA (https://www.michigan.gov/egle/contact/foia); MPART Cadillac Industrial Park AOI & FAQ, Wexford-Missaukee CTC AOI, Wexford Self-Sampling AOI (https://www.michigan.gov/pfasresponse/investigations/sites-aoi/wexford-county/); City of Cadillac fact sheet (https://www.cadillac-mi.net/DocumentCenter/View/3000/Cadillac-Water-Supply-and-Waste-Water-Treatment-Systems-Fact-Sheet), PFAS hub (https://www.cadillac-mi.net/PFAS), Laboratory page (https://www.cadillac-mi.net/186/Laboratory), DWSRF EA 2020, 2024 CCR; Wexford Co. Drain Commissioner (https://wexfordcounty.org/?page_id=469) & County Drain Map PDF; Road Commission (https://wexfordcrc.org); Equalization/GIS (https://wexfordcounty.org/?page_id=705); county FOIA policy PDF; Haring Twp (https://haringtwpmi.gov); DHD#10 records page (https://www.dhd10.org/environmental-health/well-septic/records/); Wexford Conservation District (https://www.wexfordconservationdistrict.org). **News:** Cadillac News NPDES/PFAS articles; UpNorthLive April–July 2026 flood & EGLE investigation coverage (https://upnorthlive.com/news/local/state-probe-questions-cadillac-wastewater-plant-after-flood-discharge-into-clam-river); Bridge Michigan septic-PFAS reporting; 9&10 News flood/MPART coverage. **Historical/archival:** USGS topoView (https://ngmdb.usgs.gov/topoview/viewer/); MSU RS&GIS aerial archive (https://rsgis.msu.edu/imagery-archive/information); Wexford County Historical Society aerials; AcreValue plat maps; NRCS archived soil survey manuscripts (1909 Geib). **Property records:** LoopNet & Zillow listings for 1140 Plett Rd (parcel/building details); BS&A Online (https://bsaonline.com/?uid=553); Regrid.

Prioritized Action List

1. **Pull scanned pre-2000 well logs** for T22N R9W Sections 27/28/33/34/35 at https://www.egle.state.mi.us/well-logs/ — free, immediate, biggest data gain. Check whether 1140 Plett Rd itself had a well (1979 building).
2. **Check NWI Wetlands Mapper and FEMA MSC interactively** at the coordinates (44.26464, -85.39643) — closes the two unverified-layer gaps in minutes.
3. **Open the County Drain Map PDF** and confirm whether any established drain crosses Haring Twp Section 34; call the Drain Commissioner (231-779-9115).

4. **Submit EGLE FOIA (Draft A)** — the AOI consultant reports are the single most valuable unseen dataset: they almost certainly contain measured water tables, boring logs, and flow maps within a mile of the site.
5. **Submit DHD#10 request (Draft B)** for parcel well/septic/soil-evaluation records.
6. **Export the Wellogic well set** (REST query in Section 5) and build the water-table map and two cross-sections; resolve the SWL-vs-river-elevation anomaly with LiDAR-derived river stage.
7. **Pull the USGS Haring 1 observation-well record** from api.waterdata.usgs.gov and the SIR 2004-5175 model archive from the USGS Michigan office.
8. Submit the remaining requests (Drafts C–F, I) and order 2–3 historic aerial frames (1938, ~1963, ~1980) from MSU to document pre-development conditions.

This report is a research and records-planning document, not an engineering or legal opinion. Field verification (survey-grade well elevations, on-site delineation, direct water-level measurement) is required before any interpretation here is used for design, transaction, or litigation purposes.